C12W7 · TUESDAY, MARCH 3, 2026

VIBE CODING IS DEAD

Long Live *Agentic Engineering*

PART 1: VIBE CODING IS DEAD

IT WAS THE BEST OF VIBES

Andrej Karpathy, **Feb 2025**:

"I 'Accept All' always, I don't read the diffs anymore."

Andrej Karpathy, **Feb 2026**:

"Vibe coding is passe. I'd like to mass-propose agentic engineering."

👤 **Andrej Karpathy** — founding member of OpenAI, former Sr. Director of AI at Tesla, Stanford PhD

HOW WE GOT HERE

0

MANUAL

No AI assistance

1

AUTOCOMPLETE

Tab to accept suggestions · Copilot, 2022

2

CHAT-ASSISTED

Ask questions in web chats, copy answers · ChatGPT, 2023

3

VIBE CODING

Accept All, hope it works · Karpathy, Feb 2025

4

AGENTIC ENGINEERING

Orchestrate agents with human oversight · Karpathy, Feb 2026

WHAT THE RESEARCH ACTUALLY SAYS

THE HEADLINES

THE GOOD

- **26%** more tasks completed · MIT/Microsoft, 2024
- **55%** faster task completion · Peng et al., 2023
- **21%** faster at Google · Google Internal, 2024

THE BAD

- **19% slower** for experienced devs · METR, 2025
- **1.7x** more issues in AI code · CodeRabbit, 2025
- **45%** of AI code has security flaws · Veracode, 2025

⚠️ Examined 22 studies of varying quality. All had problems. Many get quoted.

THE FINE PRINT

26% MORE TASKS

Code completion, not agentic. Only counts *compliers*. Proxy metrics (PRs, commits), not value delivered. **Build success declined** 5.5 pp.

MIT/Microsoft, 2024

55% FASTER

One JS task, single session. Upwork freelancers. CI: [21%, 89%]. No quality measurement.

Peng et al., 2023

21% AT GOOGLE

p = **0.086**. Convention is p < 0.05. Not statistically significant after adjustment. Google engineers on Google tasks.

Google Internal, 2024

19% SLOWER

N=16, experienced devs on complex codebases only. Follow-up (N=57) had *worse* methodology problems — METR published their own critique.

METR, 2025

1.7X MORE ISSUES

Own product measured the issues. "AI" PRs identified by commit messages. Deeply self-serving.

CodeRabbit, 2025

45% SECURITY FLAWS

Raw model output — no spec-driven development, no code review, no process. 100+ models on 80 isolated tasks. Worst case only.

Veracode, 2025

Companies publishing white papers that support their business have obvious conflicts of interest. But even the academic papers have had significant shortcomings.

THE VENDOR CONFLICT PROBLEM

Of the **22 major studies** on AI-assisted coding:

7	5	4
FROM AI TOOL VENDORS	FROM COMPETITORS	CLEAN ACADEMIC PROFILES

- **CodeRabbit** found "1.7x more issues" in AI code — using their own product as the measurement tool. They sell AI code review for $24/dev/month.
- **Veracode** found "45% security flaws" — no specs, no review, no process at all. They sell security scanning tools.

WHAT WE CAN CONCLUDE

1. **AI increases throughput** with varying quality
2. **Juniors may benefit more** than seniors
3. We might **overestimate AI's benefit** as individuals
4. **At big orgs**, complex existing codebases, **gains are modest**
5. **But at startups, AI is becoming the primary coder**
6. **The field is moving faster than the research.** Cherny stopped writing code by hand in Nov 2025 — 4 months ago. No study has measured mature agentic workflows.
7. **Long-term velocity is unmeasured.** AI-assisted refactoring, codebase comprehension, and onboarding should improve velocity, but we need longitudinal studies to see it.
8. **AI is an amplifier**. Good process gets better. Bad process gets worse.
9. **Confirmation bias is strong.** People see what they want to see.

AGENTIC ENGINEERING IS A SKILL TO BE LEARNED,

NOT A MAGIC WAND.

THREE RULES FOR THE BUILD PHASE

RULE 1: THE WILLISON RULE

> *"If an LLM wrote the code for you, and you reviewed it, tested it, and can explain how it works to someone else — that's not vibe coding, it's software development."*
>
> — Simon Willison

⚠️ **Class rule:** Don't deploy code you can't understand.

RULE 2: SPEC FIRST, CODE SECOND

Every credible high-output practitioner uses **spec-first development:**

BORIS CHERNY

Creator of Claude Code, Anthropic

Plan Mode → iterate until solid → one-shot it

HARPER REED

Former CTO, Obama 2012 campaign

spec.md → blueprint → execute

ADDY OSMANI

Engineering leader, Google Chrome

"Waterfall in 15 min" — spec before code

KENT BECK

Creator of TDD & Extreme Programming

TDD cycle: Red → Green → Refactor

You already have `/feature-forge`: **brief → architecture → plan → implement → review**

RULE 3: CLAUDE.MD IS YOUR TEAM'S BRAIN

WHAT IT IS

- Persistent context file in your repo root
- Claude reads it at the start of **every session**
- Your project's identity: stack, conventions, structure
- 67% of repos now use them

HOW TO USE IT

- Root CLAUDE.md: Root: under 60 lines. Project: up to ~40k tokens
- Check it into git — compounds in value
- When Claude does something wrong, **add a note**
- Template on the one-sheet

"Anytime we see Claude do something incorrectly, we add it to the CLAUDE.md."

— Boris Cherny

GIT FOR TEAMS

With **2-4 people coding**, this prevents chaos:

- **Never push to main directly.** Feature branches + PRs.
- **One feature per branch.** One AI session per branch.
- **Pull before you branch.** Commit often. Small PRs.
- **Someone reviews every PR.** Read the code, don't just check that tests pass.

⚠️ If two of you push to main at the same time without branches, **you will enter merge hell.**

PART 2: MVP SCOPING

13 TEAMS, 10 DAYS

TEAM	PRODUCT	BUILDING
4CUz	NoticeMe	Social accountability for screen time
August Fan Club	Evidence of Life	Auto-timeline from your Apple Calendar
Basket Architects	Experience Architects	Living taste profile from real-world experiences
Dream Team	Campus Study Match	See who nearby is studying the same subject
J-Combinator	—	Curated NYC event recommendations
LLMao	LLMao	Activity-based social for new social circles
Millionaires	QuestCity	Gamified real-world exploration with quests
misfitAI	Wardrobe Planner	Outfit suggestions from calendar + weather + wardrobe
Roam	Roam	Matching people on NYC activity routes
Standard Deviants	—	Qualitative/mixed-methods data analysis tool
team-friendly	—	TBD
team-frontend	FreeWeight	Strength training programming + tracking
The Birds	Circle	One-tap social coordination; see live hangouts

THE 10-DAY SPRINT

Today	Thu	Mar 10	Mar 12
DEFINE + SET UP	BUILD SPRINT	BUILD SPRINT	MVP PRESENTATIONS

Everything before this was validation. **Now you ship.**

WHAT IS AN MVP?

> *"The minimum viable product is that version of a new product which allows a team to collect the maximum amount of validated learning about customers with the least effort."*
>
> — Eric Ries, The Lean Startup

- **Minimum** = least effort. **Viable** = delivers real value. Both matter.
- Not a prototype (tests usability) — an MVP **tests the business hypothesis**
- YC's Michael Seibel: "Hold the problem tightly, hold the solution loosely"
- Reid Hoffman: "If you aren't embarrassed by the first version, you shipped too late"

YOUR MVP = FIRST AHA MOMENT

Your MVP = the shortest path to your user's **first aha moment**.

YOUR MVP

- The **ONE** core flow
- User arrives → does the thing → gets value
- Demo-able in 3 minutes

NOT YOUR MVP

- "All the features we validated"
- "Everything on our landing page"
- Settings, profiles, admin panels

💡 If you can't demo it in 5 minutes, you've scoped too much.

WHAT CAN YOU FAKE?

MARKETPLACES

Manually match supply and demand behind the scenes

ALGORITHMS

Start with the most naive version, or do it by hand.

DATA-HEAVY

Use sample or mock data

SOCIAL / MATCHING

Seed with your team, friends, classmates

"Any sufficiently advanced demo is indistinguishable from a product."

PART 3: LET'S BUILD

WORK BLOCK: TWO TRACKS

TRACK A: BUILD (MOST OF TEAM)

1. **Define your MVP scope** (10 min)
   2. What's the core flow?
   3. What are you cutting?
   4. What's faked?
   5. Who's building what?
6. **Set up dev environment**
   7. Create CLAUDE.md (template on one-sheet)
   8. Set up git branching
   9. Start first feature stub

TRACK B: DEMAND GEN (1-2 PEOPLE)

- Landing page traffic **doesn't stop**
- Check analytics: visits, signups, drop-off
- Iterate on copy if conversion is low
- Keep running ads, social, outreach

💡 Don't stop driving traffic just because you're building.

WRAP UP

ASSIGNMENT: MVP DOC

DUE THURSDAY

Submit to your team folder as `20260305/mvp_doc.md`:

- **Core flow** — user → action → value
- **Tech stack** — what you're building with
- **Team roles** — who's building what, who's on demand gen
- **What's faked** — WoZ / concierge strategies
- **Demand gen status** — what's running, what are the numbers

WHAT'S NEXT

- **Thursday (c13w7)** — Build sprint. MVP docs due. Come ready to build.
- **Tuesday Mar 10 (c14w8)** — Build sprint.
- **Thursday Mar 12 (c15w8)** — **MVP Presentations.** You demo what you built.
- **Supplemental** — Dev Workflow + Git podcast (releasing this week)

You have 10 days. The workflow is set. The one-sheet is your reference.

Go build.

END OF CLASS 12

Next: Build Sprint (Thursday)